

Automated Emerging Cyber Threat Identification and Profiling Based on Natural Language Processing

Abstract:The time window between the disclosure of a new cyber vulnerability and its use by cybercriminals has been getting smaller and smaller over time. Recent episodes, such as Log4j vulnerability, exemplifies this well. Within hours after the exploit being released, attackers started scanning the internet looking for vulnerable hosts to deploy threats like cryptocurrency miners and ransomware on vulnerable systems. Thus, it becomes imperative for the cybersecurity defense strategy to detect threats and their capabilities as early as possible to maximize success of prevention actions. Although crucial, discovering new threats is a challenging activity for security analysts due to the immense volume of data and information sources to be analyzed for signs that a threat is emerging. In this sense, we present a framework for automatic identification and profiling of emerging threats using Twitter messages as a source of events and MITRE ATT&CK as a source of knowledge for threat characterization. The framework comprises three main parts: identification of cyber threats and their names; profiling the identified threat in terms of its intentions or goals by employing two machine learning layers to filter and classify tweets; and alarm generation based on the threat's risk. The main contribution of our work is the approach to characterize or profile the identified threats in terms of its intentions or goals, providing additional context on the threat and avenues for mitigation. In our experiments the profiling stage reached a F1 score of 77% in correctly profiling discovered threats.

Reinforcement Learning Models and Algorithms for Diabetes Management

Abstract:With the advancements in reinforcement learning (RL), new variants of this artificial intelligence approach have been introduced in the literature. This has led to increased interest in using RL to address complex issues in diabetes management. Using RL, a decision maker (or agent) observes decision-making factors (or state) from the dynamic operating environment, selects actions based on its optimal policy (i.e., a mapping between states and optimal actions), and subsequently receives delayed rewards. The agent adapts its policy to changes in the operating environment to maximize its cumulative reward as time goes by, thereby improving system performance. This paper presents how various variants of RL have been used to improve diabetes management, such as a higher time in range during which the blood glucose level is within the normal range and a higher similarity between RL and physician's policies. Key highlights focus on the application of RL in diabetes management, including a taxonomy of the attributes of RL (e.g., roles and advantages), essential elements for training (e.g., data and simulators), representations of diabetes attributes in RL models, and variants of RL algorithms. In addition, this paper discusses open issues and potential future developments in the use of RL in diabetes management.

FMDADM: A Multi-Layer DDoS Attack Detection and Mitigation Framework Using Machine Learning for Stateful SDN-Based IoT Networks

Abstract: The absence of standards and the diverse nature of the Internet of Things (IoT) have made security and privacy concerns more acute. Attacks such as distributed denial of service (DDoS) are becoming increasingly widespread in IoT, and the need for ways to stop them is growing. The use of newly formed Software-Defined Networking (SDN) significantly lowers the computational burden on IoT network nodes and makes it possible to perform more security measurements. This paper proposes an SDN-based, four-module DDoS attack detection and mitigation framework for IoT networks called FMDADM. The proposed FMDADM framework comprises four main modules and five-tier architecture. The first module implements an early detection process based on the average drop rate (ADR) principle using a 32-packet window size. The second module uses a novel double-check mapping function (DCMF), that aids in earlier attack detection at the data plane level. The third module is an ML-based detection application comprising four phases: data preprocessing, feature extraction, training and testing, and classification. This module detects DDoS attacks using only seven features: two selected and five newly computed features. The last module introduces an attack mitigation process. We applied the proposed framework to three test cases: one single-node attack test case and two multi-node attack test cases, all with real IoT traffic generated and deployed in Mininet-IoT. The proposed FMDADM framework efficiently detects DDoS attacks at high and low rates, can discriminate between attack traffic and flash crowds, and protects both local and remote IoT nodes by preventing infection from propagating to the ISP level. The FMDADM outperformed most existing cutting-edge approaches across ten different evaluation criteria. According to the experimental results, FMDADM achieved the following accuracy, precision, F-measure, recall, specificity, negative predictive value, false positive rate, false detection rate, false negative rate, and average detection time benchmarks:- 99.79%, 99.43%, 99.77%, 99.79%, 99.95%, 00.21%, 00.91%, 00.23%, and 2.64 μ s, respectively.

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